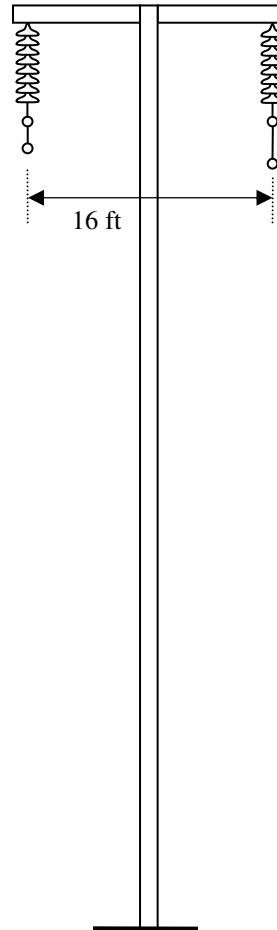


1. A 24KV, 1 ϕ transmission line uses 2 ACSR “Falcon” cables with 6” spacing in one conductor and 2 ACSR “Kiwi” cables with 10” spacing in the other conductor as shown in the support tower diagram below.

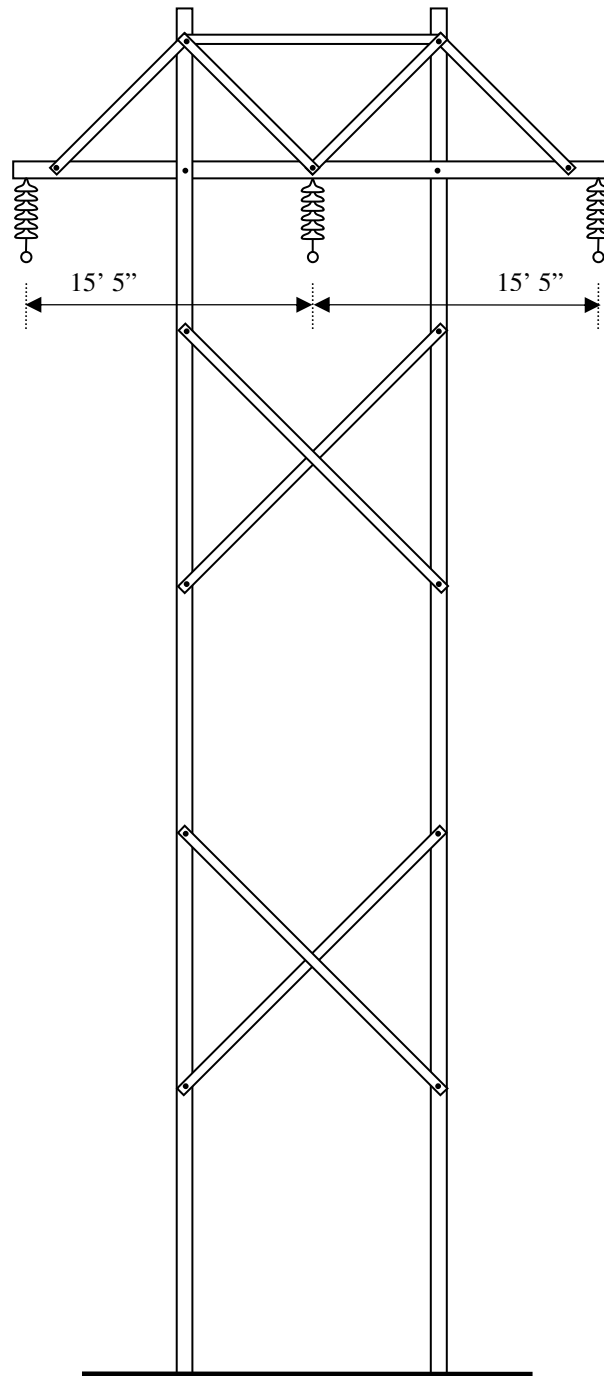


Calculate the following transmission line quantities.

- Geometric mean radius of each conductor. (0.16171ft, 0.21814ft)
 - Modified geometric mean radius of each conductor. (0.17944ft, 0.24563ft)
 - Geometric mean distance between the two conductors. (16.00824ft)
 - Distributed series resistance per mile. (0.05385 Ω /mi)
 - Distributed series resistance per mile at a temperature of 36°C. (0.05737 Ω /mi)
 - Distributed series resistance per mile at a temperature of 5°C. (0.05055 Ω /mi)
 - Distributed series inductance per mile. (2.8617mH/mi)
 - Distributed shunt capacitance per mile. (10.329nF/mi)
 - Propagation constant per mile. (0.002051 $\angle$ 88.571°/mile)
 - Characteristic impedance. (526.6859 $\angle$ -1.429° Ω)
 - Characteristic impedance at a temperature of -10°C. (526.6335 $\angle$ -1.310° Ω)
 - If the transmission line is 75 miles long, calculate the medium length model ABCD parameters. (0.988185 $\angle$ 0.034°, 81.012549 $\angle$ 87.142°, 0.00029032 $\angle$ 90.017°)
-
2. A new type of single phase insulated conductor power cable is being designed for use in commercial buildings. The cable uses two solid copper wires enclosed and separated by a special type of polyvinyl insulating material. Each conductor of the cable has a radius of 0.25" and a distributed series resistance of 0.02 Ω /1000 feet. The two conductors are separated by a center to center distance of 1". The polyvinyl insulating material used in the cable has an inductance constant that is 1.05 times that of air, and a capacitance constant that is 8.95 times that of air. Calculate the following quantities for this power cable
- Distributed series resistance per 1000 feet. (0.04 Ω /1000ft)
 - Distributed series inductance per 1000 feet. (0.20947mH/1000ft)

- c. Distributed shunt capacitance per 1000 feet. (54.737nF/1000ft)
- d. Distributed series impedance per 1000 feet. ($0.088522\angle 63.137^\circ \Omega/1000\text{ft}$)
- e. Distributed shunt admittance per 1000 feet. ($20.63536\angle 90^\circ \mu\text{S}/1000\text{ft}$)
- f. Propagation constant per 1000 feet. ($0.00135155\angle 76.568^\circ/1000\text{ft}$)
- g. Characteristic impedance. ($65.496645\angle -13.432^\circ \Omega$)

3. A 50 mile, 115KV, three phase transmission line uses an ACSR “Falcon” cable in each phase as shown in the support tower diagram below.

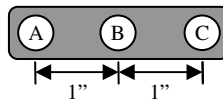


Calculate the following transmission line quantities.

- a. Geometric mean radius and modified geometric mean radius of a phase conductor. (0.0523ft, 0.0644ft)
- b. Geometric mean distance between the phase conductors. (19.42378ft)
- c. Distributed per phase series resistance per mile. ($0.0612\Omega/\text{mi}$)
- d. Distributed per phase series inductance per mile. (1.90458mH/mi)

- e. Distributed per phase shunt capacitance per mile. (15.6822nF/mi)
- f. Distributed per phase series impedance per mile. ($0.720613\angle 85.128^\circ \Omega/\text{mi}$)
- g. Distributed per phase shunt admittance per mile. ($5.91205\angle 90^\circ \mu\text{S}/\text{mi}$)
- h. Propagation constant per mile. ($0.002064\angle 87.564^\circ/\text{mi}$)
- i. Characteristic impedance. ($349.126061\angle -2.436^\circ \Omega$)
- j. Medium length model ABCD parameters. ($0.994694\angle 0.026^\circ$, $36.030670\angle 85.128^\circ$, $0.0002948\angle 90.013^\circ$)
- k. Long length model ABCD parameters. ($0.994699\angle 0.026^\circ$, $35.966976\angle 85.137^\circ$, $0.0002951\angle 90.009^\circ$)
- l. Long length model ABCD parameters at a temperature of 40°C . ($0.994699\angle 0.028^\circ$, $35.989030\angle 84.742^\circ$, $0.0002951\angle 90.009^\circ$)

-
4. A new type of three phase power cable is being designed for industrial power systems. The cable uses three solid copper wires enclosed and separated by a special type of polyvinyl insulating material as shown below. Each conductor of the cable has a radius of 0.125" and a distributed series resistance of 0.02Ω per 1000ft. The polyvinyl insulating material used in the cable has a capacitance constant that is 8.25 times greater than that of air, and has an inductance constant that is the same as that of air.



Calculate the following quantities for this power cable

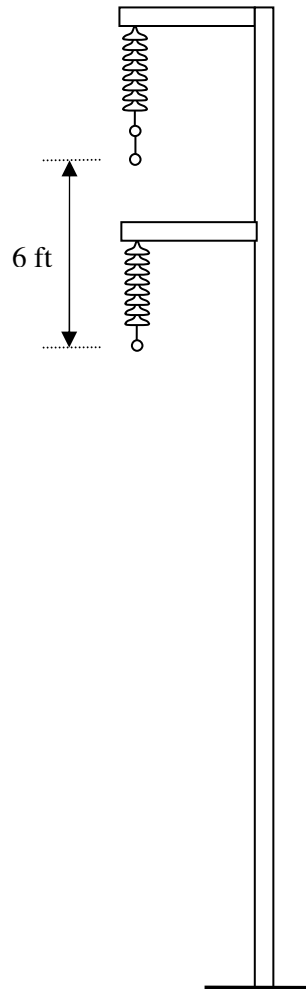
- a. Distributed per phase series resistance per 1000 feet. ($0.02\Omega/1000\text{ft}$)
- b. Distributed per phase series inductance per 1000 feet. ($0.1561\text{mH}/1000\text{ft}$)
- c. Distributed per phase shunt capacitance per 1000 feet. ($60.5471\text{nF}/1000\text{ft}$)
- d. Distributed per phase series impedance per 1000 feet. ($0.06215\angle 71.228^\circ \Omega/1000\text{ft}$)
- e. Distributed per phase shunt admittance per 1000 feet. ($22.82571\angle 90^\circ \mu\text{S}/1000\text{ft}$)
- f. Propagation constant per 1000 feet. ($0.001191\angle 80.614^\circ/1000\text{ft}$)
- g. Characteristic impedance. ($52.180343\angle -9.386^\circ \Omega$)

-
5. A special multi-purpose power cable has been designed that can be used in either single phase or three phase power system applications. The power cable is built using four individually insulated solid copper conductors enclosed by a rectangular outer jacket. Each individual conductor has a 0.25 inch radius and a resistance of 0.02Ω per mile. Each conductor also has a 0.25" layer of insulation around it, which creates a 1" center to center spacing between adjacent conductors in the cable. The insulating material used in the cable has an inductance constant that is 1.3 times greater than that of air and a capacitance constant that is 12.5 times greater than that of air. The drawing below shows a cross sectional view of this power cable.



- a. The power cable is to be used in a single phase power system application, with adjacent pairs of conductors connected together on both ends of the power cable to create two composite conductors. Calculate the characteristic impedance of the power cable for this configuration. ($53.21746\angle -1.261^\circ \Omega$)
- b. The power cable is to be used in a three phase power system application, with three of the conductors used for the ABC phases and the fourth conductor is used for neutral. If the neutral conductor is one of the outer conductors of the power cable, then calculate the characteristic impedance of the power cable for this configuration. ($33.64186\angle -1.942^\circ \Omega$)
- c. The power cable is again to be used in a three phase application, but this time the neutral conductor is one of the inner conductors of the power cable, then calculate the characteristic impedance of the power cable for this configuration. ($40.73199\angle -1.624^\circ \Omega$)

6. A 12KV, 1 ϕ transmission line uses 2 ACSR “Canary” cables with 5” spacing in one conductor and 1 ACSR “Parrot” cable in the other conductor as shown in the support tower diagram below.



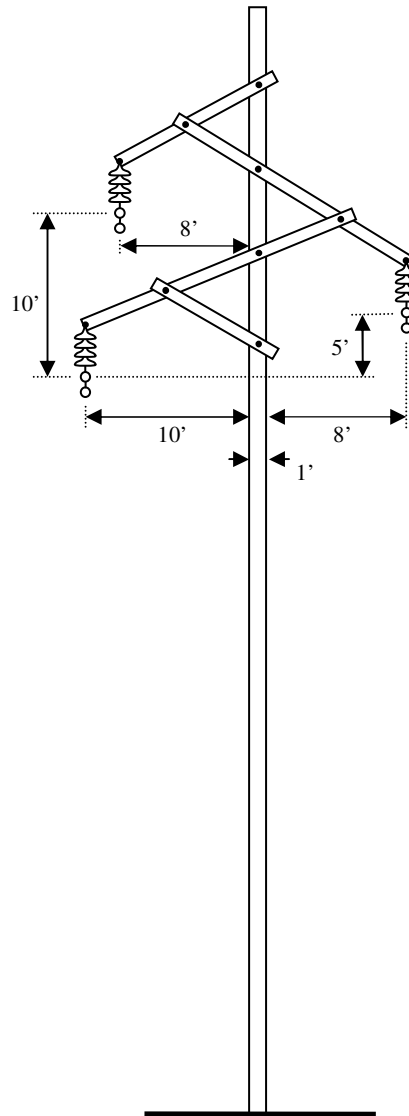
Calculate the following transmission line quantities.

- Geometric mean radius of each conductor. (0.12780ft, 0.0506ft)
 - Modified geometric mean radius of each conductor. (0.14201ft, 0.0628ft)
 - Geometric mean distance between the two conductors. (6.20484ft)
 - Distributed series resistance per mile. (0.1113 Ω /mi)
 - Distributed series inductance per mile. (2.7976mH/mi)
 - Distributed shunt capacitance per mile. (10.6963nF/mi)
 - Propagation constant per mile. (0.002068 $\angle$ 86.988 $^\circ$ /mi)
 - Characteristic impedance. (512.8349 $\angle$ -3.012 $^\circ\Omega$)
 - If the transmission line is 95 miles long, calculate the long length model ABCD parameters. (0.980872 $\angle$ 0.118 $^\circ$, 100.106743 $\angle$ 84.015 $^\circ$, 0.00038063 $\angle$ 90.039 $^\circ$)
7. A special multi-purpose power cable has been designed that can be used in either single phase or three phase power system applications. The power cable is built using six individually insulated solid copper conductors enclosed by a rectangular outer jacket. Each individual conductor has a 0.27 inch radius and a resistance of 0.0175 Ω per mile. Each conductor also has a 0.23" layer of insulation around it, which creates a 1" center to center spacing between adjacent conductors in the cable. The insulating material used in the cable has an inductance constant that is 1.35 times greater than that of air and a capacitance constant that is 13.25 times greater than that of air. The drawing below shows a cross sectional view of this power cable.

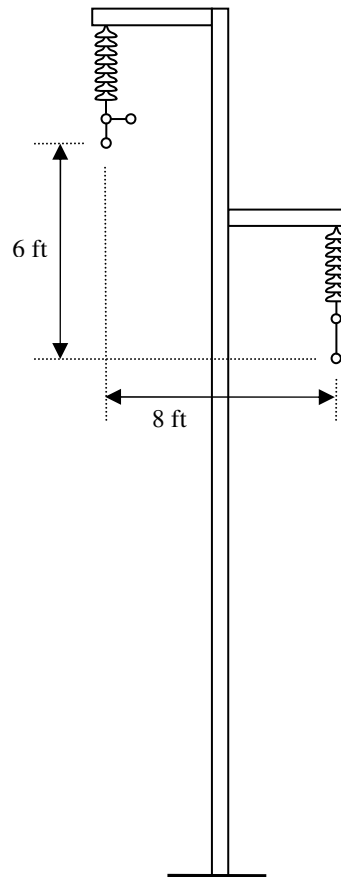


- a. The power cable is to be used in a single phase power system application, with trios of adjacent conductors connected together on both ends of the power cable to create two composite conductors. Calculate the characteristic impedance of the power cable for this configuration. ($50.938425\angle-0.743^\circ\Omega$)
- b. The power cable is to be used in a three phase power system application, with pairs of adjacent conductors used for the ABC phases. Calculate the characteristic impedance of the power cable for this configuration. ($31.398282\angle-0.898^\circ\Omega$)

8. A 75 mile, 69KV, 3 ϕ transmission line uses two ACSR *Dorking* cables per phase in 8" bundles as shown in the support tower diagram below. Calculate the following transmission line quantities.
 - a. Geometric mean radius and modified geometric mean radius of a phase conductor. (0.1166ft, 0.1322ft)
 - b. Geometric mean distance between the phase conductors. (15.25545ft)
 - c. Distributed per phase series resistance per mile. ($0.1523\Omega/\text{mi}$)
 - d. Distributed per phase series inductance per mile. ($1.56872\text{mH}/\text{mi}$)
 - e. Distributed per phase shunt capacitance per mile. ($18.8541\text{nF}/\text{mi}$)
 - f. Distributed per phase series impedance per mile. ($0.610689\angle 75.559^\circ\Omega/\text{mi}$)
 - g. Distributed per phase shunt admittance per mile. ($7.107825\angle 90^\circ\mu\text{S}/\text{mi}$)
 - h. Propagation constant per mile. ($0.002083\angle 82.779^\circ/\text{mi}$)
 - i. Characteristic impedance. ($293.117346\angle -7.221^\circ\Omega$)
 - j. Medium length model ABCD parameters. ($0.988182\angle 0.177^\circ$, $45.801641\angle 75.559^\circ$, $0.00052994\angle 90.088^\circ$)



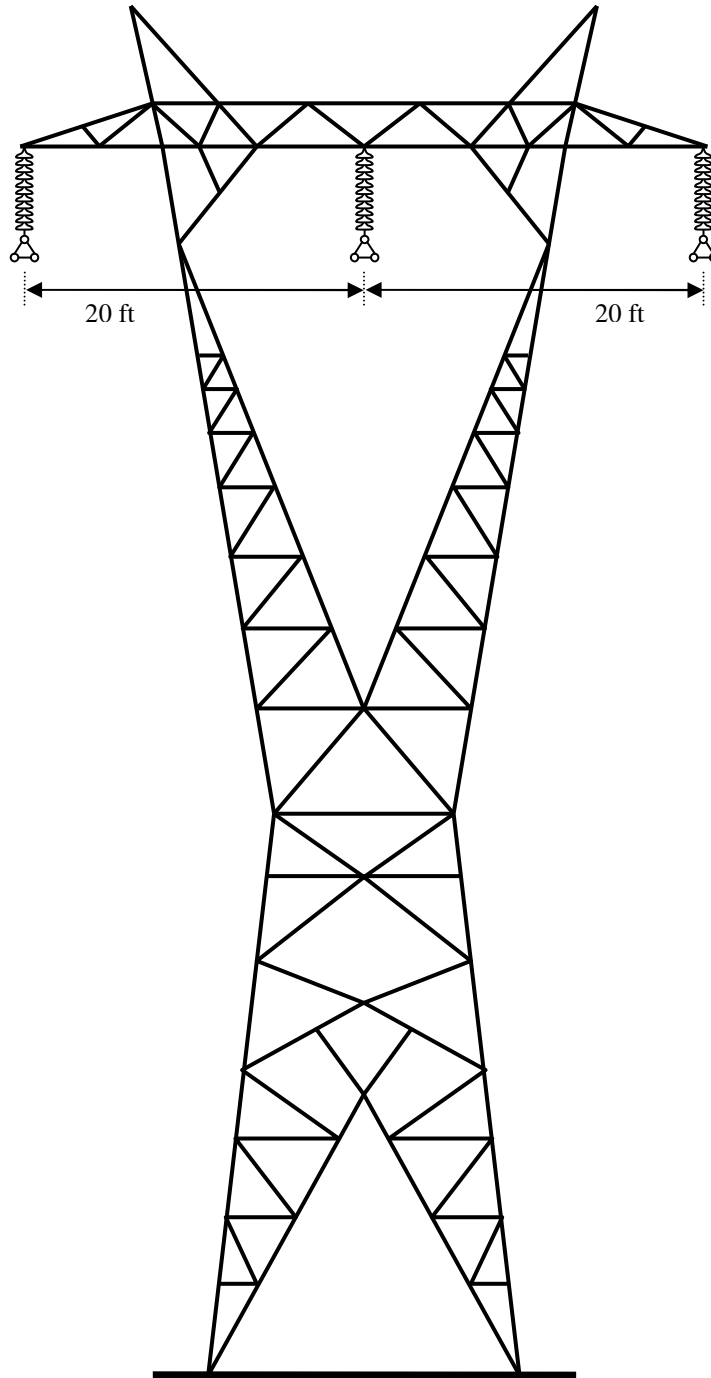
9. A single phase transmission line uses 3 ACSR “Wood Duck” cables with 4” spacing between directly adjacent cables in one conductor and 2 ACSR “Mallard” cables with 9” spacing in the other conductor as shown in the support tower diagram below.



Calculate the following transmission line quantities.

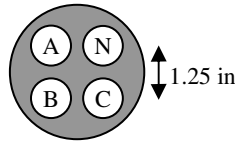
- Geometric mean radius of each conductor. (0.16854ft, 0.17146ft)
 - Modified geometric mean radius of each conductor. (0.17963ft, 0.18875ft)
 - Geometric mean distance between the two conductors. (9.82292ft)
 - Distributed series resistance per mile. (0.1069 Ω /mi)
 - Distributed series resistance per mile at a temperature of 40°C. (0.11553 Ω /mi)
 - Distributed series resistance per mile at a temperature of 10°C. (0.10258 Ω /mi)
 - Distributed series inductance per mile. (2.61145mH/mi)
 - Distributed shunt capacitance per mile. (11.2567nF/mi)
 - Propagation constant per mile. (0.00205 $\angle$ 86.901°/mi)
 - Characteristic impedance. (483.06811 $\angle$ -3.099° Ω)
 - If the transmission line is 40 miles long, calculate the medium length model ABCD parameters. (0.996658 $\angle$ 0.021°, 39.611144 $\angle$ 83.803°, 0.0001695 $\angle$ 90.010°)
-
10. A 96 mile, 230KV, 3 ϕ transmission line uses 3 ACSR “Bluebird” cables per phase in 10” equilateral bundles as shown in the support tower diagram below. Calculate the following transmission line quantities.
- Geometric mean radius and modified geometric mean radius of a phase conductor. (0.34396ft, 0.37078ft)
 - Geometric mean distance between the phase conductors. (25.19842ft)
 - Distributed per phase series resistance per mile. (0.015867 Ω /mi)
 - Distributed per phase series inductance per mile. (1.38211mH/mi)
 - Distributed per phase shunt capacitance per mile. (21.2214nF/mi)
 - Distributed per phase series impedance per mile. (0.521283 $\angle$ 88.256° Ω /mi)

- g. Distributed per phase shunt admittance per mile. ($8.00028 \angle 90^\circ \mu\text{S}/\text{mi}$)
- h. Propagation constant per mile. ($0.0020422 \angle 89.128^\circ/\text{mi}$)
- i. Characteristic impedance. ($255.26084 \angle -0.872^\circ \Omega$)
- j. Medium length model ABCD parameters. ($0.980792 \angle 0.034^\circ$, $50.043169 \angle 88.256^\circ$, $0.00076065 \angle 90.017^\circ$)
- k. Long length model ABCD parameters. ($0.980853 \angle 0.034^\circ$, $49.723369 \angle 88.267^\circ$, $0.00076312 \angle 90.011^\circ$)



11. A new type of 3ϕ power cable is being designed for industrial power systems. The cable uses four solid aluminum alloy wires enclosed and separated by a special type of polyvinyl insulating material as shown below. Each conductor of the cable has a radius of 0.5" and a distributed series resistance of $0.03\Omega/\text{kft}$. The polyvinyl insulating material used in the cable has a capacitance constant that is 12.75 times greater than that of air, and has an inductance constant that is 1.025 times greater than that of air. The four conductors

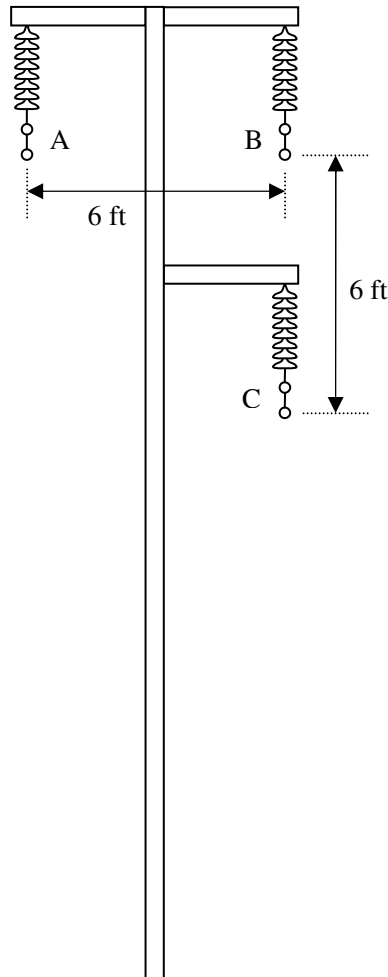
(ABCN) of the power cable are arranged inside the cable in a square configuration, with each side of the square having a conductor center to conductor center distance of 1.25".



Calculate the following quantities for this power cable

- Distributed per phase series resistance per 1000 feet. ($0.03\Omega/\text{kft}$)
- Distributed per phase series inductance per 1000 feet. ($80.09294\mu\text{H}/\text{kft}$)
- Distributed per phase shunt capacitance per 1000 feet. ($0.209533\mu\text{F}/\text{kft}$)
- Distributed per phase series impedance per 1000 feet. ($0.042564\angle 45.185^\circ\Omega/\text{kft}$)
- Distributed per phase shunt admittance per 1000 feet. ($78.99194\angle 90^\circ\mu\text{S}/\text{kft}$)
- Propagation constant per 1000 feet. ($0.0018336\angle 67.592^\circ/\text{kft}$)
- Characteristic impedance. ($23.212933\angle -22.408^\circ\Omega$)

12. A 69KV, three phase transmission line uses 2 ACSR "Osprey" cables per phase in 5" bundles as shown in the support tower diagram below.

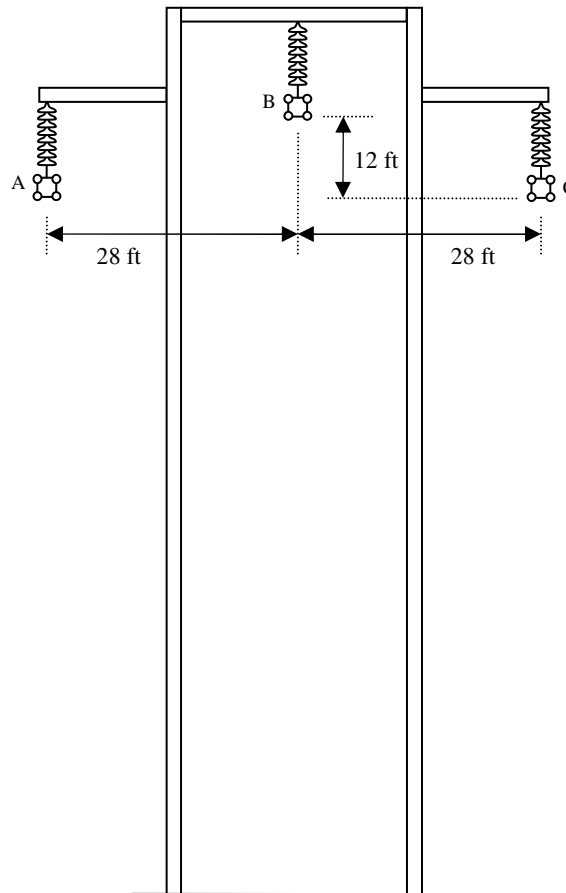


Calculate the following quantities for this transmission line.

- Geometric mean radius and modified geometric mean radius of a phase conductor. (0.1088ft, 0.1235ft)
- Geometric mean distance between the phase conductors. (6.73477ft)
- Distributed per phase series resistance per mile. ($0.08395\Omega/\text{mi}$)
- Distributed per phase series resistance per mile at a temperature of 40°C . ($0.09075\Omega/\text{mi}$)

- e. Distributed per phase series inductance per mile. (1.32794mH/mi)
- f. Distributed per phase shunt capacitance per mile. (22.3893nF/mi)
- g. Distributed per phase series impedance per mile. ($0.50761\angle 80.481^\circ \Omega/\text{mi}$)
- h. Distributed per phase shunt admittance per mile. ($8.44056\angle 90^\circ \mu\text{S}/\text{mi}$)
- i. Propagation constant per mile. ($0.0020699\angle 85.240^\circ/\text{mi}$)
- j. Characteristic impedance. ($245.23329\angle -4.760^\circ \Omega$)
- k. Medium length model ABCD parameters if the transmission line length is 20 miles. ($0.999155\angle 0.008^\circ$, $10.152192\angle 80.481^\circ$, $0.00016874\angle 90.004^\circ$)

13. A 125 mile, 345KV, 3 ϕ transmission line uses 4 ACSR “Cardinal” cables per phase in 11” square bundles as shown in the support tower diagram below.



Calculate the following transmission line quantities.

- a. Geometric mean radius and modified geometric mean radius of a phase conductor. (0.45745ft, 0.48261ft)
- b. Geometric mean distance between the phase conductors. (37.31745ft)
- c. Distributed per phase series resistance per mile. ($0.0247\Omega/\text{mi}$)
- d. Distributed per phase series resistance per mile at a temperature of -10°C . ($0.0217\Omega/\text{mi}$)
- e. Distributed per phase series inductance per mile. (1.41672mH/mi)
- f. Distributed per phase shunt capacitance per mile. (20.5914nF/mi)
- g. Distributed per phase series impedance per mile. ($0.534662\angle 87.352^\circ \Omega/\text{mi}$)
- h. Distributed per phase shunt admittance per mile. ($7.762790\angle 90^\circ \mu\text{S}/\text{mi}$)
- i. Propagation constant per mile. ($0.0020373\angle 88.676^\circ/\text{mi}$)
- j. Characteristic impedance. ($262.440464\angle -1.324^\circ \Omega$)
- k. Long length model ABCD parameters. ($0.967784\angle 0.088^\circ$, $66.113509\angle 87.381^\circ$, $0.0009599\angle 90.029^\circ$)